TITLE OF THE THESIS

TITLE OF THE THESIS

By

1. Name1x ID: xxxxxxxxx

2. Name2xx ID: xxxxxxxxx

3. Name3xxx ID: xxxxxxxxx

B.Sc. Engineering in Computer Science and Engineering

Department of Computer Science and Engineering

Faculty of Electrical and Computer Engineering

Bangladesh Army University of Science and Technology (BAUST)

OCTOBER 2023

Title of The Thesis
Title of The Thesis

This thesis is submitted in the partial fulfillment of the requirement for the degree of

B.Sc. Engineering in Computer Science and Engineering

By

1. Name1x ID: xxxxxxxxx
2. Name2xx ID: xxxxxxxxx
3. Name3xxx ID: xxxxxxxxx

Supervised by

Supervisor name

Designation, Dept. of CSE

Co-supervised by

Co-supervisor name

Designation, Dept. of CSE

Department of Computer Science and Engineering

Bangladesh Army University of Science and Technology (BAUST)

Saidpur Cantonment, Nilphamari, Bangladesh

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Board of Examiners

1. **xxxxxxxxxxxx** Supervisor
   Name of the Supervisor
   Designation: Designation of Supervisor
   Department of Computer Science & Engineering (CSE)
   Bangladesh Army University of Science & Technology (BAUST)

2. **xxxxxxxxxxxx** Member
   Name of the Member (or Co-Supervisor, if any)
   Designation: Designation of Member
   Department of Computer Science & Engineering (CSE)
   Bangladesh Army University of Science & Technology (BAUST)

3. **xxxxxxxxxxxx** Member
   Name of the Member
   Designation: Designation of Member
   Department of Computer Science & Engineering (CSE)
   Bangladesh Army University of Science & Technology (BAUST)

4. **xxxxxxxxxxxx** Member (if External)
   Name of the Supervisor
   Designation: Designation of Supervisor
   Department of Computer Science & Engineering (CSE)
   Bangladesh Army University of Science & Technology (BAUST)

.....
Name of The Head
Designation and Head
Department of Computer Science & Engineering (CSE)
Bangladesh Army University of Science & Technology (BAUST)

CANDIDATE'S DECLARATION

It is hereby declared that the contents of this project are original and any part of it has not been submitted elsewhere for the award of any degree or diploma.

Student Name	ID	Signature & Date
Student name1	20010XXXX	
Student name2	20010XXXX	
Student name3	20010XXXX	

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ABSTRACT

(250-500 words; 1 page)
This chapter provides a concise summary of your research work, outlining the main objectives, methodology, and findings of your study. The abstract is a brief, comprehensive summary of your research. It should cover the following aspects:

- The research problem or question.
- The methodology or approach used in your study.
- Key findings or results.
- Conclusions or implications.

Your abstract should be clear and concise, typically ranging from 150 to 250 words.

Keywords: Include a list of keywords that are relevant to your research. These keywords help readers and search engines identify the main topics and themes of your work. Here's an example: **Keyword 1, Keyword 2, Keyword 3,....**

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CHAPTER 1

Introduction

The introduction provides an overview of the research topic and sets the context for the study. It should include background information, the problem statement, objectives, and the significance of the research.

1.1 Background

The background section explains the context in which your research is conducted. Provide a brief history of the topic and highlight relevant previous studies and research gaps.

Table 1.1: Time Complexity for Search First Operation

Test Sample	**Time1**	**Time2**	**Time3**
100k	3.5463	3.8186	0.0041
200k	3.6123	3.7675	0.0039
300k	3.6926	3.7685	0.0042
400k	3.6699	3.7654	0.0045
500k	3.5681	3.7321	0.0038

1.2 Problem Statement

The problem statement defines the specific research problem that your thesis addresses. Clearly articulate the issue and explain why it is important to investigate.

1.3 Objectives

The objectives section outlines the goals of your research. What do you aim to achieve through your study? List the specific objectives that guide your work.

1.4 Significance of the Research

Highlight the significance of your research within the broader academic and practical context. Explain how your work adds value to the field and addresses important questions.

1.5 Organization of the Thesis

Conclude the introduction by briefly describing the organization of the thesis. Mention what readers can expect in each chapter.

Figure 1.1: Logo of Bangladesh Army University of Science and Technology.

CHAPTER 2

Literature Review

In this chapter, we review the relevant literature related to the topic of your thesis or research. It is essential to provide a comprehensive overview of existing research, theories, and studies that relate to your work.

2.1 Section Title 1

Start by discussing key concepts and theories that are relevant to your research. Provide an in-depth analysis of these concepts and how they relate to your study. Cite relevant sources using the `cite` command [1].

2.1.1 Subsection Title

You can use subsections to further organize your literature review. Discuss specific studies, methodologies, and findings that are important for your research. Make sure to include critical analysis and comparisons between different works.

2.2 Section Title 2

Continue by exploring more literature related to your research. Highlight any gaps or limitations in the existing literature that your research aims to address. Discuss the importance of your work in filling these gaps.

2.2.1 Subsection Title

Provide a detailed overview of specific research methods and approaches that have been used in previous studies. Explain how these methods have contributed to the understanding of the topic.

2.3 Summary

Summarize the key points discussed in this literature review chapter. Emphasize the significance of your research in the context of the existing literature.

CHAPTER 3

Methodology

In this chapter, we describe the research methodology employed to achieve the objectives of this study. The methodology encompasses the research design, data collection methods, data analysis techniques, and ethical considerations.

3.1 Research Design

Our research follows a [quantitative/qualitative/mixed-methods] research design. This design was chosen because it is well-suited to the research questions and objectives of this study.
Data Flow diagram:

3.1.1 Sampling

The study utilizes a [random/purposive/stratified] sampling technique to select participants. The sample size is determined using [sampling formula/justification], resulting in [number] participants.

3.1.2 Data Collection

Data is collected through [surveys/interviews/observations/questionnaires/document analysis]. The data collection instruments are designed to elicit [type of data] from the participants. We ensure the reliability and validity of our instruments through [pilot testing/pretesting].
For the list of symbols

3.1.3 Data Analysis

Data analysis is conducted using [statistical software/qualitative analysis software]. For quantitative data, we employ [descriptive statistics/inferential statistics] to [analyze patterns/test hypotheses]. For qualitative data, we use [thematic analysis/content analysis] to [identify themes/extract insights].

3.2 Ethical Considerations

This study adheres to ethical guidelines and principles. Participants are provided with informed consent forms, and their anonymity and confidentiality are guaranteed. The research is conducted with respect for the rights and dignity of the participants.

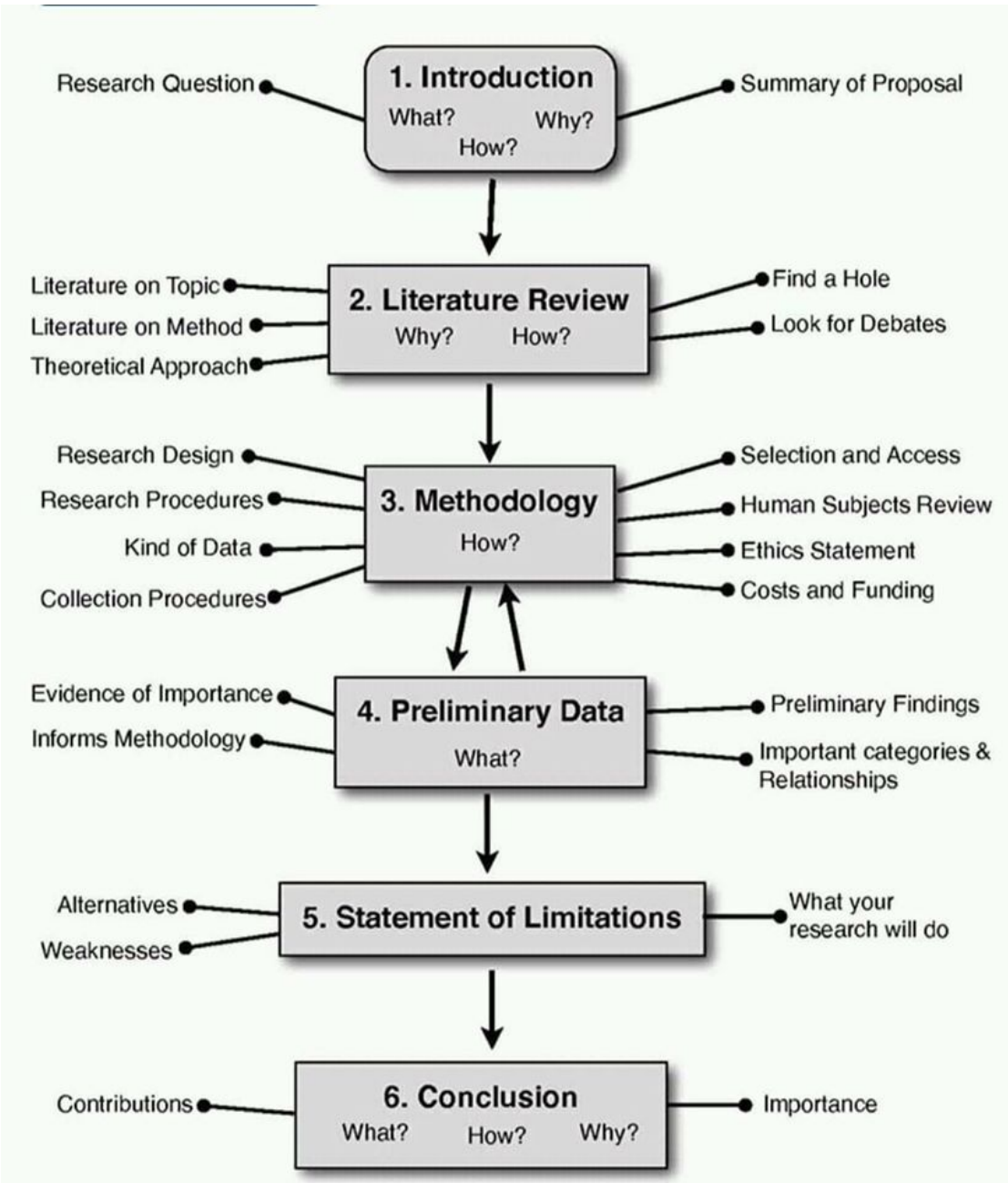

Figure 3.1: Research Article Organisation

3.3 Limitations

It is important to acknowledge the limitations of our methodology. [Discuss potential limitations, such as sample size, data collection constraints, and any other relevant issues.]

3.4 Conclusion

The methodology described in this chapter serves as the framework for conducting our research. It enables us to systematically collect and analyze data to address the research questions and achieve our objectives.

In the next chapter, we will present the results of our research and discuss their implications in the context of the existing literature.

CHAPTER 4

Implementation

4.1 System Architecture

Describe the overall architecture of the system you implemented. Include diagrams, flowcharts, or any visual aids that help the reader understand the system's structure.

4.2 Technologies Used

List the programming languages, libraries, and tools you used for the implementation. Provide versions if relevant.

4.3 Data Collection and Preprocessing

Explain how you collected and prepared the data, if applicable. Describe any data cleaning, transformation, or feature engineering processes.

4.4 Algorithm/Methodology

Explain the core algorithm or methodology you used. Provide equations, pseudocode, or detailed descriptions of the steps involved.

4.5 Code Organization

Discuss how your code is organized. Explain the directory structure and the purpose of each file or module.

4.6 Implementation Details

Provide specific implementation details, including code snippets, if relevant. Discuss any challenges you encountered during implementation and how you resolved them.

4.7 Testing and Validation

Explain how you tested and validated your implementation. Include details about the datasets used for testing and the evaluation metrics.

4.8 Performance Results

Present the performance results of your implementation. Use tables, charts, and graphs to visualize the outcomes. Discuss any insights or findings from the results.

4.9 Deployment and User Guide

If applicable, describe how to deploy the system or use the software. Provide a user guide or instructions for running the code.

4.10 Summary

Summarize the key points of the implementation chapter and how it contributes to the overall project or research.

CHAPTER 5

Result and Analysis

In this chapter, we will analyze and interpret the results obtained from the experiments conducted in the previous chapter. The analysis will be organized into sections based on the research questions and objectives outlined in Chapter. We will also present relevant statistical tests, visualizations, and discussions to provide a comprehensive understanding of the results.

5.1 Research Question 1: [Your Research Question 1]

5.1.1 Descriptive Statistics

Begin by providing an overview of the data collected and any relevant descriptive statistics. This may include means, standard deviations, medians, and other measures of central tendency and dispersion.

Table 5.1: Sample Table

Header 1	**Header 2**	**Header 3**
Row 1, Column 1	Row 1, Column 2	Row 1, Column 3
Row 2, Column 1	Row 2, Column 2	Row 2, Column 3
Row 3, Column 1	Row 3, Column 2	Row 3, Column 3

5.1.2 Hypothesis Testing

If applicable, present the results of hypothesis testing using appropriate statistical tests. Include the test statistics, p-values, and a discussion of whether the null hypotheses were rejected or not.

5.1.3 Discussion

Interpret the results of Research Question 1, considering the implications of the findings in the context of your research objectives. Discuss any unexpected outcomes or trends.

5.2 Research Question 2: [Your Research Question 2]

5.2.1 Descriptive Statistics

Provide a similar analysis as in Section 5.1, but focus on the data and statistics related to Research Question 2.

5.2.2 Hypothesis Testing

If applicable, conduct and present the results of hypothesis testing related to Research Question 2.

5.2.3 Discussion

Interpret the results of Research Question 2, exploring how they relate to your research goals and any connections to the findings in previous sections.

5.3 Overall Discussion

Summarize the key findings from your research questions and their implications. Discuss any limitations of the study and potential areas for future research.

5.4 Conclusion

Conclude the result analysis chapter by summarizing the main takeaways from your analysis and how they contribute to the broader research field.

CHAPTER 6

Conclusion and Future Work

6.1 Conclusions

6.2 Future Recommendation

6.3 Limitations/Conflict of Interests

REFERENCES

[1] M. S. Hema, R. Maheshprabhu, M. Nageswara Guptha, P. A. Mary G, and A. Sharma, "Prediction of parkinson disease using autoencoder convolutional neural networks," in *2022 International Interdisciplinary Humanitarian Conference for Sustainability (IIHC)*, 2022, pp. 236–239.

APPENDIX A

Place your source code here (if needed)/others, etc.

APPENDIX B

BIOGRAPHY

of

Full Name

Photo

Brief biography (Name, Parents name, SSC, HSC, BSc, Cultural club, Special Achievements, Merit list, Scholarship, Research Interest, Publications, etc.)

.....

Rohul Amin Labid
+8801611187733
labiddbzg@gmail.com
Jamurki, Mirzapur, Tangail

BIOGRAPHY

of

Full Name

Photo

Brief biography (Name, Parents name, SSC, HSC, BSc, Cultural club, Special Achievements, Merit list, Scholarship, Research Interest, Publications, etc.)

.....
Name
Phone
Email
Permanent Address

BIOGRAPHY

of

Full Name

Photo

Brief biography (Name, Parents name, SSC, HSC, BSc, Cultural club, Special Achievements, Merit list, Scholarship, Research Interest, Publications, etc.)

.....
Name
Phone
Email
Permanent Address